Obsidian Capital – Decentralized Capital Markets

Financial Mathematics & Quantitative Systems Reference 2025

Prepared by:
Jonathan B. Samuel, Chief Executive Officer & Chief Investment Officer
Ramesh Gopal, Chief Technology Officer

1 Bond Pricing and Yield Dynamics

Formula

$$P(y) = \sum_{t=1}^{T} \frac{c}{(1+y)^t} + \frac{F}{(1+y)^T}$$

Explanation

Determines the present value of a bond as the discounted stream of coupon payments and face value using yield y. This is the foundation for measuring yield spreads and arbitrage opportunities.

Use Case

Forms the base yield curve for sovereign and corporate bonds traded across BRICS+, European, and Middle Eastern exchanges.

2 FX Conversion and Effective Yield Multipliers

Formula

$$P_i^{USD} = P_i \times FX_i$$

Explanation

Converts local bond or equity prices into USD to maintain a single valuation basis across markets.

Use Case

Used for cross-border comparisons of yield and to normalize returns across BSE, Bursa Malaysia, Tirana Stock Exchange, and Tawadul.

3 Global Arbitrage Profit Metric

Formula

$$\Pi_{A\to B}^{practical} = \left(\frac{(1+r_A^{eff})^{IN}}{(1+r_B^{eff})^{OUT}} \right) \frac{P_B^{USD}(1-h_B^{liq})}{P_A^{USD}(1-h_A^{liq})} - 1 - tc - s - f$$

Explanation

Calculates the net profit of transferring capital between two markets after accounting for liquidity haircuts, transaction costs, systemic spreads, and fees.

Use Case

Executed by the DRTCO module to evaluate when synthetic credit issuance between jurisdictions yields net positive arbitrage.

4 Multi-Market Cycle Arbitrage

Formula

$$\Pi_{i\to j\to k\to i}^{cycle} = (1+\Pi_{i\to j})(1+\Pi_{j\to k})(1+\Pi_{k\to i}) - 1$$

Explanation

Measures compounded returns from sequential arbitrage across three or more markets.

Use Case

Models rotational capital strategies among Asia–Europe–Gulf corridors.

5 Stochastic Monte Carlo Model

Formulas

$$dX_t = \mu\, dt + \Sigma^{1/2} dW_t$$

$$X_{t+\Delta t} = X_t + \mu \Delta t + \Sigma^{1/2} \sqrt{\Delta t}\; Z$$

Explanation

Represents asset evolution through time with drift μ and covariance Σ. The discrete form drives Monte Carlo simulations.

Use Case

Used to stress-test OICD treasury reserves and arbitrage portfolios under random market shocks.

6 Synthetic Credit Spread

Formula

$$s = r_f + \theta \sigma_I^2$$

Explanation

Defines the spread above the risk-free rate as a function of credit volatility.
θ = risk-aversion parameter.

Use Case

Supports DRTCO synthetic bond pricing and treasury risk modeling.

7 Liquidity Haircut Model

Formula

$$h_i^{liq} = \alpha \frac{AvgDailyNotional}{MarketDepth} + \beta\, BidAskSpread$$

Explanation

Quantifies market-specific liquidity discounts based on trading depth and spread.

Use Case

Applied to portfolio weighting and to adjust arbitrage profitability.

8 OICD Peg Function

Formula

$$P_{OICD} = \frac{\sum_i w_i(r_i - \sigma_i^2/2)}{\sum_j v_j(LIBOR_j + CPI_j)} \times FX_{Basket}$$

Explanation

Defines the algorithmic peg for the Ozhumanill Infrastructure Crown Dollar (OICD) as a weighted average of adjusted asset returns over reference rate baskets.

Use Case

Maintains equilibrium between real-asset yields and fiat benchmarks, ensuring peg stability.

9 DRTCO Reserve Yield Function

Formula

$$R_{DRTCO} = \frac{\sum_i A_i r_i^{adj}}{\sum_i A_i}$$

Explanation

Computes the weighted adjusted yield of all assets in the DRTCO reserve pool.

Use Case

Used to rebalance treasury holdings and forecast reserve profitability.

10 Infrastructure ROI and Capital Efficiency

Formula

$$ROI_{sector} = \frac{NetInfrastructureReturn}{CapitalDeployed \times (1 + \lambda_{risk})}$$

Explanation

Assesses capital efficiency of infrastructure investments, adjusted for sectoral risk.

Use Case

Prioritizes infrastructure allocations in energy, logistics, and digital sectors.

11 System Flow Overview (diagram description)

Visual Design: Navy-silver-white arrows showing capital and data flows.

```
OICD Reserve Engine
        |
        ▼
DRTCO Treasury System
        |
        ▼
Arbitrage & Synthetic Yield Engine
        |
        ▼
Infrastructure Investment Portfolio
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Explanation

Each layer feeds performance and liquidity data upward:

- *OICD* supplies currency stability.
- *DRTCO* manages yield reserves and synthetic credit.
- *Arbitrage Engine* identifies profit differentials.
- *Infrastructure Portfolio* channels profits into tangible assets.

12 Integration of Formulas

All formulas are interoperable: arbitrage profits influence synthetic credit spreads; Monte Carlo simulations stress-test OICD reserves; liquidity haircuts feed into ROI adjustments.
Together they form a **quantitative feedback loop** aligning trading performance with long-term infrastructure value creation.

13 Appendix – LaTeX Sources

Plain-text renderings of each equation for developer use in Jupyter or Solidity modules.
(List repeats equations 1 – 10 in LaTeX syntax.)